Wifi Module 1

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1. **In which OSI layer does the Wi-Fi standard/protocol fit?**

Wi-Fi works mainly in two layers of the OSI model: the Physical Layer (Layer 1) and the Data Link Layer (Layer 2). At Layer 1, Wi-Fi handles the physical transmission of data, like how signals are sent over the air using radio waves. At Layer 2, it manages how the data is packaged and addressed for communication between devices. This includes things like using MAC addresses to ensure that data goes to the correct device.

2. **Can you share the Wi-Fi devices that you are using day-to-day life, share that device's wireless capability/properties after connecting to the network. Match your device to corresponding Wi-Fi Generations based on properties.**

For example, if I'm using a MacBook Pro, it connects to Wi-Fi using 802.11ac (which is Wi-Fi 5). When it connects to the network, it typically works on both the 2.4 GHz and 5 GHz bands and can reach speeds of up to 1.3 Gbps (on the 5 GHz band). It uses advanced techniques like 256-QAM (a way to pack more data into signals) and can handle 80 MHz or 160 MHz channels for faster speeds. This setup places the device in the Wi-Fi 5 generation, which is known for offering good performance, especially on the 5 GHz band.

3. **What are BSS and ESS?**

A BSS (Basic Service Set) is a simple network made up of a single Access Point (AP) and the devices connected to it. The BSSID is the unique identifier for that specific AP. An ESS (Extended Service Set) is a larger network made up of multiple BSSs that are connected, often via a wired network. The SSID is the name that identifies an entire ESS. ESS allows users to roam between different APs, which is useful in bigger areas like offices or homes.

4. **What are the basic functionalities of a Wi-Fi Access Point?**

A Wi-Fi Access Point (AP) acts as a central hub in a wireless network. It connects wireless devices to a wired network, so you can use your device to access the internet or share files. The AP sends and receives data signals over the air, usually on the 2.4 GHz or 5 GHz frequency bands. It also manages things like traffic between devices, security (like encryption), and supports features like roaming, allowing you to move between different APs without losing connection.

5. **Difference between Bridge mode and Repeater mode.**

In Bridge mode, the AP connects two separate network segments. It typically links a wireless network to a wired one without changing the IP addresses of devices. Repeater mode is different because the AP picks up the signal from a main AP and rebroadcasts it, extending the network's range. While Bridge mode connects networks together, Repeater mode simply extends the reach of an existing network but may reduce the speed because the AP is retransmitting data.

6. **What are the differences between 802.11a and 802.11b?**

802.11a operates on the 5 GHz band and can provide speeds up to 54 Mbps. It uses OFDM (Orthogonal Frequency Division Multiplexing) to improve performance and is less affected by interference but has a shorter range because higher frequencies don't travel as far. 802.11b, on the other hand, works on the 2.4 GHz band and offers lower speeds, up to 11 Mbps. It uses a simpler method called DSSS (Direct Sequence Spread Spectrum) and has a longer range but is more likely to experience interference from devices like Bluetooth and microwaves.

7. **Configure your modem/hotspot to operate only in 2.4 GHz and connect your laptop/Wi-Fi device, and capture the capability/properties in your Wi-Fi device. Repeat the same in 5 GHz and tabulate all the differences you observed during this.**

When using the 2.4 GHz band, the connection speed typically ranges from 72 Mbps to 150 Mbps. The range is good, but there's more interference from other devices like microwaves, which also use the 2.4 GHz band. When switching to 5 GHz, the speed increases significantly, often up to 1 Gbps or more, with less interference from other devices. However, the 5 GHz signal doesn't travel as far and has trouble passing through walls compared to 2.4 GHz. The 5 GHz band also allows for wider channels, offering faster speeds, but with less range.

8. **What is the difference between IEEE and WFA?**

IEEE (Institute of Electrical and Electronics Engineers) is an organization that creates the standards for how technologies like Wi-Fi work. The IEEE 802.11 standard is the blueprint for how Wi-Fi operates. On the other hand, the Wi-Fi Alliance (WFA) is a group that certifies Wi-Fi products, ensuring that they meet these standards and can work with other Wi-Fi devices. The Wi-Fi Alliance also manages the Wi-Fi logo, meaning that certified devices have been tested for compatibility and reliability.

9. **List down the type of Wi-Fi internet connectivity backhaul, share your home/college's wireless internet connectivity backhaul name and its properties.**

Wi-Fi backhaul refers to how the network connects to the internet or other parts of the network. Some common types are:

Wired Ethernet backhaul: The most common and reliable, using Ethernet cables to connect APs to the internet.

Wireless Mesh backhaul: APs connect wirelessly to each other, extending coverage without needing wires.
Fiber-optic backhaul: Uses high-speed fiber-optic cables to provide extremely fast internet, typically used in commercial or large-scale networks.
In my home setup, we have fiber-optic backhaul connected to a Wi-Fi 6 router, which provides fast, stable internet across the house, supporting both 2.4 GHz and 5 GHz bands.

10. List down the Wi-Fi topologies and use cases of each one.

There are different ways to set up Wi-Fi networks, depending on the needs:

Ad-Hoc: This is a simple setup where devices communicate directly without an AP. It's used for temporary connections, like sharing files between two devices.
Star Network: The most common setup, where devices connect to a central AP, which connects to the internet or a wired network. It's used in homes, businesses, and public spaces.
Mesh Network: This setup uses multiple APs that connect wirelessly, allowing seamless coverage over a large area. It's perfect for large buildings or outdoor areas.
Point-to-Point (P2P): This setup is used to connect two locations over a long distance using specialized equipment like directional antennas. It's common for linking buildings or areas that are far apart.